

Frédéric Andrian

AI Engineer · Agentic Systems and Platform Tooling
Bordeaux, France

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Professional Summary

AI Engineer completing an MSc-equivalent engineering degree in Artificial Intelligence and Cognitive Science. At INRIA (French National Institute for Research in Digital Science and Technology), I independently developed an end-to-end multi-step agent without an orchestration framework to understand and control its tools, state, memory, context, and execution. I also built Python and FastAPI services, RAG components, Dockerized tools, GitHub CI/CD, and rigorous LLM evaluation protocols. This experience strengthened my interest in the infrastructure and tooling required to operate reliable agents. I now want to continue toward agentic platform engineering while expanding my cloud and DevOps expertise. Available from November 2026 and motivated to grow within an international environment.

Experience

INRIA (French National Institute for Research in Digital Science and Technology), Mnémosyne Team Bordeaux, France

- AI Research Engineer Intern Mar. 2026 – Sep. 2026
- Independently designed and developed, with guidance from my supervisors, an end-to-end multi-step agent that helps researchers discover scientific papers.
 - Built the agent without an orchestration framework to understand and control tool execution, state, memory, context, and multi-step behavior.
 - Integrated the OpenAI API and designed the tools used by the agent to retrieve information and complete successive tasks.
 - Built Python and FastAPI backend services, REST APIs documented with OpenAPI, and MongoDB persistence.
 - Implemented a RAG pipeline using document chunking, embeddings, semantic search, and ChromaDB.
 - Dockerized selected components and implemented a GitHub CI/CD pipeline to automate integration.
 - Managed a Dockerized web application for LLMs installed locally on a Mac Studio, including model installation and configuration.
 - Evaluated OpenClaw in an isolated Docker environment with attention to the reliability and security of autonomous tool and file access.
 - Benchmarked Qwen, Gemma, and GLM on an HPC cluster with Slurm across latency, TTFT, response quality, and context size.
 - Evaluated multiple quantization configurations to identify the best trade-off between inference performance and response quality for each execution environment.
 - Defined an experimental protocol, analyzed the results, and contributed to a scientific manuscript documenting the model evaluation.
 - Developed a Python tool to automatically collect scientific paper metadata from multiple public sources.
 - Implemented webhooks to relay the results of scheduled tasks.
 - Developed a React and Next.js interface that enabled researchers to interact with the agent and its tools.

Karunya University Coimbatore, India

- Reinforcement Learning Research Intern Jun. 2025 – Sep. 2025
- Designed and compared deterministic path planning with a reinforcement learning approach for autonomous navigation of a hospital robot.
 - Deployed and optimized the software and reinforcement learning model on an embedded Raspberry Pi.

Technical Skills

Agentic Systems	Framework-free multi-step agents, tool design, state, memory and context management, OpenAI API, OpenClaw
LLM Evaluation	Qwen, Gemma, GLM, quantization, experimental protocols, latency, TTFT, response quality, context size
Backend and Integration	Python, FastAPI, Node.js, REST APIs, OpenAPI, webhooks
RAG and Data	Chunking, embeddings, semantic search, ChromaDB, MongoDB, SQL
DevOps and Deployment	Linux, Docker, Git/GitHub, CI/CD, Slurm, HPC environments, Raspberry Pi
Frontend	React, Next.js
Machine Learning	PyTorch, Deep Learning, Reinforcement Learning, NLP
Languages	French (native), English (professional, IELTS 7.5), Spanish (basic)

Education

ENSC, Bordeaux INP Bordeaux, France

- Engineering Degree, MSc-equivalent, in Artificial Intelligence and Cognitive Science 2023 – 2026
- Coursework in software engineering, machine learning, large language models, natural language processing, deep learning, and human-computer interaction.